CYTECH ACADEMY

Reverse Engineering

Complete Study Notes - beginner friendly, exam and lab ready.

(c) CyTech. For enrolled students. Unauthorized attack is illegal - practice only in labs.

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Foundations

Reverse engineering means understanding a program without its source code. You need: x86/x64 assembly basics (registers EAX/EBX/ECX/EDX, stack, CALL/RET, JMP/JE/JNE), the compilation pipeline (C -> compiler -> assembler -> linker -> PE/ELF binary), and how memory is laid out (stack, heap, .text, .data).

Binaries come in formats: PE (Windows .exe/.dll), ELF (Linux), Mach-O (macOS). Learn to read headers with PE-bear, CFF Explorer, readelf and objdump.

Static Analysis

Static = without running the file. Start with: file, strings (find URLs, passwords, messages), hashes (VirusTotal lookup), PE/ELF headers, imported DLLs/functions (Imports reveal capability: CreateRemoteThread = injection, InternetOpen = network).

Disassemblers/decompilers: Ghidra (free, NSA), IDA Free, Binary Ninja (commercial), radare2/cutter. Ghidra's decompiler view turns assembly into near-C pseudocode - read that first.

Dynamic Analysis

Dynamic = run it and watch. Debuggers: x64dbg (Windows), gdb with gef/pwndbg (Linux). Set breakpoints on interesting API calls, step through, watch registers and memory change.

Behavioral monitoring: Process Monitor (file/registry), Process Explorer, Wireshark (network), FakeNet-NG (simulate internet in a safe VM), Regshot. Always analyze malware in an isolated VM with no shared folders and a host-only network (FLARE-VM or REMnux).

Malware Analysis Workflow

1) Triage: hash, strings, AV detections. 2) Static: unpack if packed (UPX is trivial: upx -d; packed malware hides imports). 3) Dynamic: what files dropped, what registry keys, what C2 domains. 4) Write indicators of compromise (IOCs): hashes, domains, mutexes, file paths.

Packing/obfuscation is the biggest beginner wall: look for high-entropy sections, few imports (LoadLibrary+GetProcAddress only), and a small stub that unpacks the real payload into memory at runtime.

Crackmes & Practice

Learn safely with crackmes (crackmes.one): small legal puzzles where you find the correct serial/password by reading assembly. Beginner path: patch a JNE to JE, then derive the real serial algorithmically.

Then progress to CTF reversing challenges (PicoCTF, HTB) and simple ransomware samples shared for education on MalwareBazaar (VM only, never on your host).